

Ground Truth: Deterministic Falsification of Authority Continuity in Autonomous Systems

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The reviewed Ground Truth v1 freeze is now remotely inspectable in the public repository `groundtruthexhibit-beep/ground-truth-exhibit`. The freeze documentation commit is `3d09acc804bc1271965696f2d9561143ca28b835` and records frozen research HEAD `31d78aa158618ca8b72eb61e2f49b434e4319e9e`. The frozen contract is `authority-lab-v1` with six tuple coordinates, six invariants, and three outcomes; the corpus is `LAB-V0-001..012` plus `LAB-V1-013..466` (466 fixtures). The freeze records 260/260 public tests PASS on the freeze branch and 466/466 Authority Lab fixtures PASS at the frozen research HEAD. These results remain bounded failure-to-falsify evidence under the exact stated model, corpus, implementation, assumptions, and repository states; they are not a claim of universal security or formal proof.

Abstract

Autonomous systems can act through chains of human approvals, delegated authority, policy decisions, identity bindings, credentials, connectors, tools, persistent state, and execution backends. Each component can appear individually valid while the composite execution no longer has sufficient evidence of authority at the moment its effects occur. This paper studies that failure mode as authority continuity. The central analytical separation is Capability != Evidence != Authority != Execution: technical ability to act, possession of evidence, justified authority, and actual execution are distinct properties.

Ground Truth is an executable deterministic falsification framework for testing authority continuity under a frozen public model, authority-lab-v1, with six tuple coordinates, six invariants, and three terminal outcomes: AUTHORIZED, DENIED, and UNAVAILABLE. Rather than treating a successful authorization result as evidence of correctness, Authority Lab constructs reproducible cases intended to expose executions that remain AUTHORIZED despite missing, stale, replayed, transferred, rolled-back, conflicting, compositionally invalid, or event-misaligned authority evidence.

The methodology follows counterexample construction, deterministic reproduction, normative classification, hostile design review, implementation, adversarial regression testing, positive controls, and exact-state hostile review. Investigated classes include objective binding, mutation closure, observation completeness, successor continuity, replay and equivocation, verifier rollback, source independence, delegation laundering, human-approval continuity, execution-target continuity, and temporal composition.

A principal temporal finding is that locally current authority heads can still refer to different execution events. This motivates common-event convergence: when a protected effect consumes authority, every authority-bearing component required by policy must establish applicability to the same protected execution event. Ground Truth does not claim universal security or formal proof. PASS means only that no reproducible in-scope counterexample was found under the stated model, fixture corpus, assumptions, implementation, and exact evaluated repository state.

Reviewed freeze checkpoint: repository groundtruthexhibit-beep/ground-truth-exhibit; freeze branch docs/ground-truth-research-freeze-v1; freeze documentation commit 3d09acc804bc1271965696f2d9561143ca28b835; frozen research HEAD 31d78aa158618ca8b72eb61e2f49b434e4319e9e; schema authority-lab-v1; 260/260 public tests PASS on the freeze branch; 466/466 Authority Lab fixtures PASS at the frozen research HEAD; frozen corpus LAB-V0-001..012 and LAB-V1-013..466. The exact reviewed Git objects are remotely inspectable.

Keywords: autonomous systems; authorization; authority continuity; delegation; provenance; deterministic testing; falsification; zero trust; usage control; agentic systems.

Core thesis: Autonomous systems can preserve the machinery required to perform an action without preserving sufficient evidence that the action remains authorized at the event where its effects occur.

Capability != Evidence != Authority != Execution

Authority continuity is not persistence of authorization; it is the persistence or re-establishment of the evidence required to justify authorization.

1. Introduction

Autonomous systems need not exercise authority through a single contemporaneous decision. A consequential action can depend on human approval created earlier, delegated authority received from another agent, identity established by a control plane, policy evaluated against observed state, a credential accepted by a connector, and a terminal execution performed later against an external resource. Each step can appear individually legitimate while the complete execution is not.

Such systems often preserve capability more easily than justification. Credentials survive process restarts. Tokens are cached. Delegations are serialized. Approvals persist in databases. Jobs wait in queues. Connectors are rebound. Tool calls are retried. Workers are replaced. Policy results are memoized. The continued existence of those mechanisms does not establish that the evidence supporting their earlier use still applies.

Ground Truth studies this gap as authority continuity. Authority is treated not as a durable property owned by an agent, credential, approval, or token, but as a relation that must remain justified by evidence applicable to the protected execution. A decision valid for one subject, artifact, identity basis, boundary epoch, target, or execution event cannot simply be inherited by another state because the system retains the ability to act.

This perspective is deliberately narrower than general AI safety. Ground Truth does not determine whether an objective is desirable, whether an agent is intelligent, or whether a policy is socially optimal. It asks a systems-security question: does the evidence admitted by the stated authority model justify the authority claimed for the protected effect that is about to occur?

The approach is falsification-oriented. Authority Lab seeks reproducible cases in which the evaluated system returns `AUTHORIZED` even though the required authority evidence should instead yield `DENIED` or `UNAVAILABLE`. One valid in-scope counterexample is sufficient to falsify the behavior under test; failure to find a counterexample does not prove correctness.

The most important composite result motivating this manuscript is temporal. Different authority subsystems can each maintain an internally current local head while those heads refer to different points in execution history. An approval valid through event 2 and a delegation valid through event 2 can still remain present when a protected effect occurs at event 3. If the system checks only local currentness, historical authority can be composed into present execution authority.

Local currentness does not imply globally current authority.

Ground Truth therefore develops the idea of a canonical authority-consuming execution event. Common-event convergence does not require every subsystem to recompute authorization continuously. It requires that, when a protected effect consumes authority, the evidence required by policy be shown to apply to that same event.

1.1 Contributions

1. An authority-continuity formulation for failures in which capability, state, credentials, or previous authorization persist while the evidence needed to justify continued authority does not.
2. A strict analytical separation between capability, evidence, authority, and execution.
3. An executable three-outcome model that preserves `AUTHORIZED`, `DENIED`, and `UNAVAILABLE` rather than collapsing uncertainty into a Boolean decision.
4. A deterministic falsification workflow based on counterexamples, minimized reproducers, hostile review, repair, regression fixtures, positive controls, and exact-state checkpoints.
5. A taxonomy of authority-continuity counterexamples spanning objective binding, mutation and observation completeness, restart/successor continuity, replay, rollback, provenance, delegation, human approval, execution targets, and temporal composition.
6. A synthesis of recurring repair patterns: canonical binding, closure, lineage, non-rollbackable authority-relevant history, provenance, fail-closed uncertainty, effect identity, boundary binding, anti-replay, durable consumption, and common-event convergence.
7. A temporal-composition requirement: authority fragments required for a protected effect must converge on the same authority-consuming execution event.
8. A reproducibility discipline that binds scientific claims to exact repository state and preserves historical vulnerable behavior rather than reporting only the final passing suite.

2. Problem Statement

The central problem is that persistence of authorization-related state does not necessarily imply persistence of justified authority. An autonomous system can retain capability, credentials, a prior decision, a human approval, a delegation record, identity state, control-plane state, and tool access while no longer possessing sufficient evidence that those facts jointly authorize the execution now being attempted.

Ground Truth uses authority continuity to describe the preservation or re-establishment of the evidence required to justify authority across authority-relevant transitions leading to a protected execution.

2.1 Authority Is Relational

Within the frozen public model, an authority decision is represented over the tuple:

(subject, artifact, control_state, identity_basis, boundary_epoch, decision)

The six frozen invariants are INV-CONT, INV-DISC, INV-BND, INV-EVD, INV-USE, and INV-CLO. The following definitions are reproduced verbatim from the canonical Layer 2 contract.

INV-CONT — verification-to-commit continuity

The exact live tuple used by the irreversible transition must be shown continuous with the tuple evaluated for authority. The exact live objective_binding, including its objective, decision contract, authority source, scope, lineage, generation, currentness, revocation, and supersession conditions, must also remain continuous or be re-established through T3. A T1 binding does not survive a relevant T2 change merely because its record or label remains available. Continuity must hold through the commit point, not merely at request receipt, evidence admission, validation, or effect preparation. A change or unresolvable gap in any required relation before commitment yields UNAVAILABLE, unless an established applicable condition instead yields DENIED.

INV-DISC — identity-bearing discriminator discipline

Identity-bearing values must be established by the applicable identity_basis. Objective, decision-contract, objective-binding, policy, schema, authority-source, wrapper, derivation, and transformation identities and versions must be discriminated wherever they are authority relevant. A path, name, label, role, handle, tag, replica name, PID, UID, session identifier, address, semantic-similarity claim, or byte equality may select or compare a candidate but does not alone prove identity or fidelity. The same decision-contract bytes under different objectives do not share an objective binding. Replacement, ABA reuse, rollback, restoration, cloning, aliasing, and same-content substitution cannot inherit authority merely through a stable selector or equal bytes.

INV-BND — independent epoch-bound boundary

The applicable authority boundary must be independently established and bound to a non-restorable authority lineage expressed by boundary_epoch. The source of an objective_binding and any delegation must belong to that applicable boundary and current authority lineage. The requester cannot choose, replace, route around, bootstrap, or collude into a favorable boundary, binding source, verifier, registry, or semantic interpreter. No objective binding is inherited implicitly across organizations, tenants, environments, sandbox and production, authority domains, boundary epochs, or authority roots. Restart, fork, restore, replica divergence, promotion, rejoin, verifier replacement, authority rotation, or another authority-relevant lineage event must be distinguishable unless continuity is independently established.

INV-EVD — exact evidence binding and non-promotion

Evidence is bound to the exact tuple and decision for which it was admitted. Evidence may satisfy an authority decision only when it satisfies the exact admitted decision contract and an exact current authoritative objective_binding admits that contract for the applicable authority objective and exact decision context. Evidence is not authority. Evidence for proposition P', correct verifier execution, provenance, identity, signatures, proofs, repetition, aggregation, transformation, or consensus cannot establish the objective binding or promote an unadmitted contract into authority. An objective binding admits the contract but does not itself establish operational evidence required by that contract. Evidence cannot be silently replayed across a subject, artifact, control state, identity basis, boundary epoch, or decision, and weak evidence cannot become stronger through relabeling, repetition, aggregation, transformation, or consensus. Missing required evidence binding, objective binding, or evaluation yields UNAVAILABLE.

INV-USE — commit-time consumption discipline

A one-shot authority instance authorizes at most one committed authority-consuming transition. A continuing grant

and its objective binding remain usable only while their exact current validity conditions hold. An expired, revoked, superseded, out-of-scope, wrong-objective, wrong-contract, wrong-effect, wrong-generation, wrong-boundary, or otherwise ineligible binding is not usable. Cached success, cached binding, retry, restart, rollback, failover, residue, duplicate delivery, or absence of a consumption or revocation record cannot recreate authority. Unknown prior consumption or unknown current binding status does not become fresh authority; it yields UNAVAILABLE.

INV-CLO — no implicit transformation or aggregate closure

Authority over an artifact does not implicitly authorize a composition, partition, encoding, wrapper, projection, report, derivative, batch, aggregate, or other transformed object. An objective binding for decision contract D does not automatically admit a wrapped, compiled, optimized, translated, migrated, summarized, derived, or equivalent-looking D2. Each distinct authority-bearing result and decision contract requires its own exact current binding unless an exact current authoritative relationship explicitly admits that exact transformation and resulting contract for the same objective, scope, and boundary context. Transformation success or an equivalence proof is evidence for the proposition it establishes; it does not create authority. Separate valid decisions do not imply authority for their combined outcome.

A valid authorization result should therefore be interpreted as authority for a specific authority context, not as a transferable license to continue acting. Changes in subject, artifact, identity basis, boundary epoch, or evidence applicability can invalidate continuity even when execution machinery is unchanged.

2.2 Authority-Continuity Failure

An authority-continuity failure occurs when a system preserves or reconstructs enough authority-related state to return AUTHORIZED even though the evidence required by the stated model no longer justifies that result. Causes include staleness, replay, omitted mutation, incomplete observation, successor substitution, rollback, false source independence, delegation expansion, approval transfer, target substitution, and event mismatch.

2.3 Local Currentness and Global Authority

A human-approval subsystem can consider an approval current relative to its own interval. A delegation subsystem can consider a grant current relative to its own transition history. A connector can consider a credential current. A verifier can consider its local state current. These statements can all be locally correct without jointly establishing that the same evidence authorizes the same terminal execution.

In the evaluated temporal model, suppose approval and delegation are each valid only through event 2 while a protected effect occurs at event 3. If authorization tests only whether each local head is internally current, the system can combine historically valid evidence into a globally stale authorization. The defect is therefore a join problem as well as a freshness problem.

2.4 Canonical Authority Consumption

The reviewed T3 repair uses one protected canonical T3_EXECUTION_EVENT as the common authority-consuming join point. Human approval, delegation, tool/connector state, policy, successor/boundary state, verifier state, anchors, witnesses, control-plane evidence, and the terminal effect bind or otherwise demonstrate applicability to that event where policy requires them.

Canonical authority consumption is narrower than continuous authorization. It does not require every subsystem to recompute authority at all times; it requires the evidence consumed by a protected effect to establish applicability to the same execution event.

The temporal result progressed through reproduced counterexample, reviewed design, implementation, adversarial regression, positive AUTHORIZED reachability, and exact-state hostile review. Common-event convergence is therefore reported as a tested authority-continuity condition within the frozen model, not as a universal proof that every real-world effect admits a single sufficient execution-event representation.

2.5 Research Questions

- RQ1 - Falsification: Can deterministic adversarial construction expose cases in which a system returns **AUTHORIZED** even though authority evidence should be **DENIED** or **UNAVAILABLE**?
- RQ2 - Repair without loss of reachability: Can those cases be repaired without making legitimate **AUTHORIZED** paths unreachable?
- RQ3 - Temporal/compositional convergence: When authority-bearing subsystems advance independently, what evidence is necessary to show that their locally current states jointly authorize the same terminal execution event?

2.6 Scope of a Ground Truth Result

A Ground Truth **FAIL** identifies a reproducible in-scope case in which the evaluated system does not satisfy the expected authority behavior of the stated model. A **PASS** means only that no reproducible in-scope counterexample was found under the stated model, fixture corpus, assumptions, implementation, and exact evaluated repository state.

PASS does not imply universal system security, fixture completeness, absence of undiscovered counterexamples, correctness of external evidence, formal proof unless separately established, or safety outside the stated authority model.

The falsification logic is asymmetric: one valid counterexample can disprove a claimed in-scope behavior; failure to find a counterexample does not prove that none exists.

3. Threat Model

Ground Truth evaluates failures in the preservation and composition of authority evidence. The threat model includes deliberate adversarial manipulation and ordinary distributed-system behavior capable of producing the same authority error. The protected property is:

A terminal protected execution must not be classified **AUTHORIZED unless the evidence required by the stated authority model justifies authority for that execution.**

The adversary succeeds by causing the system to preserve, reconstruct, replay, transfer, substitute, compose, or misinterpret authority-related state such that the final decision becomes **AUTHORIZED** despite insufficient applicable authority evidence.

3.1 Adversary Capabilities

- Replay previously valid evidence or move it across subjects, objectives, boundaries, targets, or execution events.
- Exploit incomplete mutation tracking or incomplete observation admission.
- Create stale-successor or ABA histories across shutdown and restart.
- Replay or equivocate observation commitments.
- Exploit verifier rollback or split-brain state.
- Present correlated evidence as independent evidence.
- Launder delegated authority across agents or expand scope during re-delegation.
- Transfer human approval to a different effect, actor, target, or time.
- Substitute tool connectors, backends, credentials, accounts, or permissions.
- Cause locally current authority components to refer to different execution events.

3.2 Non-Malicious Failure Modes

The same conditions can arise without a malicious attacker through delayed execution, retries, worker reassignment, service restart, stale caches, queue persistence, asynchronous approval workflows, race conditions, replication, failover, rollback, connector rebinding, credential refresh, or long-running agent handoff. The relevant property is

not whether the triggering condition is adversarial or accidental, but whether authority remains justified when that condition occurs.

3.3 Protected Decision and Supporting Evidence

The protected object of analysis is the authority decision governing a terminal protected execution. Supporting evidence can include subject identity, artifact or objective identity, control state, identity basis, boundary epoch, grant identity, approval identity, effect identity, connector or tool target, verifier state, anchors, witnesses, control-plane relationships, and temporal execution context.

Not all of these are tuple coordinates in the frozen public model. Some are evidence used to establish or validate the frozen coordinates. The paper preserves this distinction and does not silently expand the six-coordinate contract.

3.4 Four Evidence Categories

Category	Meaning
Experimental evidence	Behavior directly demonstrated by Authority Lab at an identified repository state.
Normative requirement	What the stated model or a reviewed design says must hold.
Repair evidence	Experimental evidence associated with a specific implemented repair and identified repository state.
External assumption	A fact the laboratory accepts but cannot independently establish.

3.5 Consistent-Lie Boundary

Authority Lab reasons over admitted evidence. If every authority-relevant admitted source presents a mutually consistent but false account of the external world, and no contradictory or independently grounded evidence survives into the model, the lab may have no internal basis for detecting the deception. This limitation follows from the fact that Authority Lab evaluates admitted evidence rather than directly observing external reality.

The boundary is especially important for claims involving witnesses, anchors, provenance, source independence, and control-plane collusion. The model can detect explicitly represented common roots; it cannot necessarily detect hidden collusion that leaves no observable evidence within the model.

3.6 Trusted Computing Base and External Trust

The final release should distinguish the implementation trusted computing base from external trust assumptions. The implementation trusted computing base can include canonical serialization, fixture parsing, oracle logic, cryptographic implementations used by the harness, repository contents, and the execution environment. External trust assumptions can include source identifiers, authority roots, external identity facts, clocks, or observations accepted as axiomatic inputs.

Implementation TCB: fixture loader and schema validation; canonical serialization and domain-separated SHA-256 digest construction; the deterministic oracle and reason-code logic; protected commitment/anchor manifests used by the harness; test runner and Python execution environment; and the exact repository contents under review. External trust assumptions: truthfulness of admitted external identity/signature facts, authority-root facts, clocks and observations; integrity of sources outside the modeled repository; survival of at least one contradictory or independently current source when rollback/collusion detection is required; and correctness of physical effects outside the modeled protected execution event. If all admitted sources lie consistently, all comparison roots are compromised or rolled back together, and no contradictory independent evidence survives, Authority Lab cannot infer omitted external reality.

3.7 Out of Scope

- Universal autonomous-system correctness or safety.
- Correctness of arbitrary natural-language objectives.
- Confidentiality or general-purpose intrusion detection.
- Physical-world truth independent of admitted evidence.

- Absence of all hidden collusion.
- Correctness of external human decisions.
- Formal verification of the complete system unless separately established.
- Correctness beyond the frozen authority model.

3.8 Adversary Success Condition

For a given experiment, the adversary succeeds when a protected execution is classified AUTHORIZED while the stated model requires DENIED or cannot establish sufficient evidence and therefore requires UNAVAILABLE, and that discrepancy is reproducible under the stated experimental conditions.

Observed outcome = AUTHORIZED, while required outcome is DENIED or UNAVAILABLE.

3.9 Repair Success Condition

A repair is not adequate merely because the original counterexample stops returning AUTHORIZED. A mature repair should eliminate the reproduced counterexample, preserve the correct DENIED-versus-UNAVAILABLE semantics, retain legitimate AUTHORIZED positive controls, survive adversarial variants, preserve the frozen 6/6/3 contract unless an explicit reviewed model revision is made, reproduce deterministically, identify the exact repository state, and survive exact-state hostile review.

4. Ground Truth Authority Model

The public schema is authority-lab-v1. It deliberately fixes six tuple coordinates, six invariants, and three outcomes. Later mechanisms such as grant identity, approval identity, verifier state, witness provenance, execution-target identity, or execution-event identity may be evidence used to establish the frozen model; they do not silently create additional tuple coordinates.

Coordinate	Role
subject	The entity whose authority is evaluated.
artifact	The protected object, objective, or authority-relevant target.
control_state	Authority-relevant state required to decide the exact transition, including commit/effect/ordering/consumption facts when applicable; omitted required state yields UNAVAILABLE rather than an implicit new coordinate.
identity_basis	The basis on which subject identity is established.
boundary_epoch	The authority-relevant structural boundary context.
decision	AUTHORIZED, DENIED, or UNAVAILABLE.

4.1 Outcomes

AUTHORIZED means that the authority requirements of the stated model are satisfied by admitted evidence for the evaluated state. DENIED means that available evidence positively establishes that the requested authority should not be granted. UNAVAILABLE means that the evidence necessary to establish the requested authority decision safely is missing, stale, ambiguous, conflicting, insufficiently bound, unverifiable, or otherwise outside the conditions required by the model.

DENIED != UNAVAILABLE

4.2 Evidence Is Not the Tuple

Human approvals, grants, commitments, successor relationships, verifier checkpoints, witness statements, anchors, connectors, credentials, and execution events may be required evidence without becoming new tuple coordinates. This distinction keeps the public model frozen while allowing evidence discipline to become stricter.

4.3 Authority Consumption

A decision can be evaluated at one point and consumed later. A stored AUTHORIZED result is therefore not sufficient by itself to establish later execution authority. The evidence dependencies that justified the earlier decision must remain applicable or be re-established at the authority-consuming event.

4.4 Local Heads and Event Applicability

Composite systems can maintain local authority heads for approval, delegation, policy, tools, verifier state, anchors, witnesses, and control planes. Each head can be locally valid while the set is globally stale. The model therefore distinguishes local currentness from applicability to the protected execution.

Local currentness does not imply globally current authority.

5. Authority Lab

Authority Lab is an executable deterministic falsification harness. Its primary experimental object is a canonical JSON fixture representing authority-relevant state, admitted evidence, mutation history, and expected behavior. In the frozen v1 research state the loader is `authority_lab/runner.py::load_fixture`, the oracle is `authority_lab/oracle.py::evaluate`, canonical digest construction is `authority_lab/oracle.py::_canonical_digest`, and fixtures reside under `cases/authority_lab`. Protected commitment and anchor state are represented by `trusted_commitment_state.json` and `trusted_anchor_state.json` where applicable.

Canonicalization uses deterministic JSON serialization for the modeled fields and domain-separated SHA-256 digests. The fixture schema is `authority-lab-v1`. The runner rejects malformed fixture state rather than allowing fixture expectations to redefine oracle-owned relationships. The frozen release corpus is LAB-V0-001..012 plus LAB-V1-013..466. Exact schema details and field requirements are recorded in the frozen fixture/canonicalization documentation.

5.1 Deterministic Oracle

The oracle independently evaluates authority evidence and returns AUTHORIZED, DENIED, or UNAVAILABLE. Determinism matters because it makes failures transferable, repair effects attributable, and disagreement inspectable. The frozen v1 release includes explicit repeated-run evidence over identical canonical input.

Determinism evidence: `AuthorityLabTests.test_identical_canonical_input_replays_deterministically` evaluates the same protected composite-positive input five times and requires identical verdict/reason output. Oracle-independence evidence includes `AuthorityLabTests.test_wrong_expectation_does_not_change_actual`, `AuthorityLabTests.test_expected_outcome_remains_outside_v1_oracle_input`, and `AuthorityLabTests.test_labels_titles_and_expectation_do_not_change_oracle`. A targeted oracle-independence/determinism run passed 3/3; the frozen public suite passed 260/260.

5.2 Oracle Independence

Fixture expectations do not determine the oracle verdict. The oracle fixes the actual authority outcome before comparing it with the fixture expectation. A deliberately incorrect expected answer leaves the oracle result unchanged and produces `CASE_MISMATCH`. This separates an oracle decision from a test expectation and makes disagreement evidence rather than a mechanism for changing the result.

5.3 Positive, Negative, and Epistemic Controls

Control	Expected outcome	Purpose
Positive	AUTHORIZED	Shows legitimate authority remains reachable.
Negative	DENIED	Shows evidence can positively establish impermissibility.
Epistemic	UNAVAILABLE	Shows uncertainty is not promoted to permission.

5.4 Exact-State Review

Repository state is part of experimental identity. A hostile review tied to one exact commit does not automatically transfer to a later commit. Publication evidence should preserve repository, immutable commit or release identifier, relevant parent where meaningful, clean worktree state, commands, test output, and hostile-review result.

6. Falsification Methodology

The experimental progression is: valid baseline -> authority-relevant mutation -> candidate counterexample -> deterministic reproducer -> normative classification -> repair design -> hostile design review -> implementation -> regression -> positive control -> exact-state checkpoint.

6.1 Start Valid, Then Mutate Authority

Where possible, investigation begins from a legitimate AUTHORIZED path. A minimal mutation changes an authority-relevant dependency while preserving unrelated state. High-value mutations preserve enough superficial continuity that a weak implementation is tempted to reuse earlier authority.

6.2 Preserve Original Observed Behavior

When a mutated fixture remains AUTHORIZED despite a required DENIED or UNAVAILABLE result, the original permissive behavior is preserved as evidence. It is not rewritten away by changing fixture expectations. The historical state explains why the repair exists and what regression would recreate.

6.3 Separate Fixture Defect, Oracle Defect, and Model Defect

An unexpected result is not automatically an implementation counterexample. The investigation must establish that the fixture is well formed, the mutation is in scope, the normative result follows from the stated model or an explicit reviewed requirement, and the observed behavior is reproducible.

6.4 Repair and Attack the Repair

A repair is derived from the broken authority assumption, not from the literal fixture bytes. After implementation, the original counterexample becomes a permanent regression fixture and related variants attempt to bypass the repair. A valid positive path must remain AUTHORIZED.

7. Counterexample Classes

The investigated classes are analytical families with explicit evidence maturity. At the reviewed v1 freeze, objective binding, mutation-registry closure, observation completeness, shutdown/successor continuity, commitment replay/equivocation, verifier rollback/split-brain, source independence, control-plane independence, delegation continuity, human-approval continuity, tool/execution-target continuity, T3 execution-event convergence, and Composite adversarial continuity each reached Level 7 under the paper's evidence ladder. Level 7 is an evidence-maturity state, not a security rating.

Class	Representative failure	Repair direction
Objective binding	Authority for one objective or artifact is reused for another.	Canonical objective/artifact binding.
Mutation-registry closure	An authority-relevant transition occurs outside the observed mutation surface.	Closure or safe UNAVAILABLE.
Observation completeness	An incomplete observation is treated as complete state.	Completeness/admission evidence.
Shutdown/successor ABA	Visible identity returns after a discontinuity and inherits old authority.	Successor lineage and boundary epoch.
Commitment replay/equivocation	Authentic commitments are replayed or conflicting states coexist.	Freshness, uniqueness, anti-replay.
Verifier rollback/split-brain	Authority history moves backward or diverges.	Non-rollbackable durable history.
Anchor/source independence	Multiple facts share one stale or compromised root.	Provenance-aware independence.
Control-plane false independence	Separate controls share hidden control or collusion.	Dependency-aware control-plane evidence.
Delegation laundering	Downstream authority exceeds or disconnects	Grant lineage, scope preservation, event

Class	Representative failure	Repair direction
	from its root grant.	applicability.
Human approval continuity	Real approval is replayed, transferred, or consumed by another effect.	Approval-to-effect/event binding.
Tool/execution-target continuity	A logical tool remains authorized while connector, backend, credential, or target changes.	Canonical terminal target binding.
Temporal composition	Locally current authority heads refer to different execution events.	Common-event convergence.

7.1 Cross-Class Pattern

Across these classes, a state transition occurs after evidence was established. The system retains the evidence and the capability to act, then assumes the evidence still applies. The transition may be semantic, temporal, identity-related, boundary-related, observational, infrastructural, or compositional. The recurring failure is inherited authority without demonstrated continuity.

8. Repair Classes

Authority-continuity counterexamples repeatedly expose a smaller set of architectural problems. The repair classes below are not universal security laws. They are patterns derived from reproduced failures and reviewed design responses. A repair pattern is experimentally supported only where corresponding implementation, regression fixtures, positive controls, and exact-state review exist; otherwise it remains a normative design requirement.

Authority continuity is not persistence of authorization; it is the persistence or re-establishment of the evidence required to justify authorization.

Repair class	Core requirement
Canonical binding	Bind authority to the canonical subject, artifact, grant, effect, target, or event identity that matters.
Closure	Prevent authority-relevant transitions from occurring outside the evidence surface while old authority remains consumable.
Explicit lineage	Preserve auditable origin across delegation, succession, restart, and derived state.
Non-rollbackable authority-relevant history	Prevent disappearance of history from recreating obsolete authority; this does not require all state to be numerically monotonic.
Provenance-aware independence	Treat independence as an evidence property, not a count of interfaces or messages.
Fail-closed uncertainty	Missing required evidence yields UNAVAILABLE rather than permission.
Exact effect binding	Bind upstream authority to the terminal effect that consumes it.
Boundary/epoch binding	Require continuity evidence across structural discontinuities.
Anti-replay binding	Distinguish authenticity from current applicability.
Durable consumption	Preserve the fact that one-use authority has already been consumed.
Common-event convergence	Require all mandatory authority evidence to apply to the same protected execution event.
Composite repair	Evaluate whether individually correct controls are jointly applicable to the same execution.

8.1 Canonical Binding

Authority should bind to the exact object whose identity matters to the decision. Weak designs often bind authority to a proxy - for example, approval to a workflow when the relevant requirement is approval of a protected effect, or tool authorization to a logical tool name when the authority-sensitive target is the resolved execution target.

General principle: if two states should not share authority, they must not collapse to the same authority binding. Binding is only useful if the bound identifier itself is trustworthy and stable; a perfectly bound false identity remains false evidence.

8.2 Closure Over Authority-Relevant Mutation

If an unobserved transition can change whether authority remains valid, the authority model is not closed over its relevant mutation surface. An authority-relevant transition must either be represented in the admitted evidence surface or force a state in which authority cannot safely be established. Closure is stronger than logging: old authority must not remain consumable across an invisible authority-relevant transition.

Closure is always relative to a stated model. Ground Truth does not prove that all physically possible authority-relevant mutations have been modeled; it tests whether the declared mutation surface behaves consistently.

8.3 Explicit Lineage

Authority that survives transformation must preserve evidence of where it came from. This is especially important across delegation, succession, restart, re-entry, derived artifacts, control-plane transitions, and authority-root changes. A terminal agent should not inherit authority solely because the final edge of a chain is authentic; the lineage must justify the resulting terminal authority.

Lineage establishes continuity only relative to the roots and transitions accepted by the model. If the root itself is false, compromised, or misidentified, lineage alone cannot recover external truth.

8.4 Non-Rollbackable Authority-Relevant History

The requirement is not that all authority state be numerically monotonic. Revocations, replacements, supersession, policy changes, and boundary transitions can change semantics. The requirement is that authority-relevant history whose disappearance would recreate obsolete authority cannot silently move backward.

Examples include consumed approvals, replay-prevention state, verifier history, revocations, successor epochs, commitment sequence state, and durable checkpoints. Ground Truth does not prescribe one universal persistence mechanism; the semantic requirement is that authority-invalidating history must not disappear in a way that recreates prior authority.

8.5 Provenance and Source Independence

Multiplicity does not establish independence. Several authority signals can still reduce to one effective source if they share an authority root, control plane, signer, witness lineage, acquisition path, or rollback domain. Where policy depends on independence, evidence must include enough provenance to evaluate that independence.

Ground Truth cannot detect hidden shared control if the dependency is absent from admitted evidence. This remains an explicit residual trust boundary.

8.6 Fail-Closed Uncertainty

MissingRequiredEvidence => UNAVAILABLE

Ground Truth rejects the inference `NoProofOfDenial => AUTHORIZED`. Missing, stale, conflicting, ambiguous, or otherwise insufficient authority evidence must not be converted into permission. Preserving `UNAVAILABLE` avoids manufacturing authority from uncertainty while also distinguishing uncertainty from positive denial.

8.7 Exact Effect Binding

Human approvals, delegations, policies, and tool authorizations often describe intent at a higher level than the terminal effect. If planning, connector resolution, backend selection, or resource mutation changes authority-relevant meaning, upstream authority must remain bound to the actual protected effect rather than merely to an intermediate abstraction.

8.8 Boundary and Epoch Binding

Structural discontinuities such as restart, successor replacement, trust-domain transition, re-entry, authority-root change, or control-plane replacement must not be treated as ordinary continuation. Authority crossing such a discontinuity requires evidence that continuity remains valid across the boundary. A correct repair must still permit legitimate successor transitions; otherwise it degenerates into blanket refusal.

8.9 Anti-Replay Binding

Authenticity answers whether evidence was genuinely produced; replay resistance answers whether that evidence remains applicable now. Replay-resistant authority evidence can require binding to an epoch, nonce, sequence, objective, subject, effect, validity interval, or execution event. The mechanism can vary; the property is that previously consumed or expired evidence cannot later be accepted as fresh authority.

8.10 Durable Consumption State

Some authority objects are consumable. Once a one-time approval, delegation, nonce, replay token, or execution authorization is consumed, that fact becomes authority-relevant state. If a restart or rollback forgets consumption, stale evidence can become usable again. Durable consumption therefore depends on preserving the fact of consumption across all transitions relevant to later authorization.

8.11 Common-Event Convergence

Common-event convergence is the key repair pattern emerging from temporal composition. Local freshness is insufficient when authority-bearing components refer to different execution histories. The stronger requirement is that every authority component required for a protected effect establish applicability to the same authority-consuming event.

AUTHORIZED(T) => for every required H_i: Current(H_i) AND Applies(H_i, T)

Local currentness does not imply globally current authority.

Common-event convergence differs from continuous authorization. It does not require every subsystem to continuously recompute authority. It requires that, when a protected effect is about to consume authority, the evidence required by policy be shown to apply to the same authority-consuming execution event.

T3 evidence status: counterexample reproduced; design reviewed; implementation completed at 4e1eae61136fdda078c5e38f71e0f62f83fa92f9 (parent 74fba1f8fd537afb59996662affe1df64457d530); 411 prior fixtures migrated with zero prior expected-outcome changes; LAB-V1-412..443 added; 443/443 Authority Lab fixtures and 257/257 public tests passed; focused exact-head tests passed 7/7. Exact hostile verdict: PASS — NO REPRODUCIBLE IN-SCOPE CROSS-LAYER TEMPORAL-COMPOSITION OR STALE-EXECUTION BYPASS FOUND.

8.12 Composite Repair

Individually repaired authority modules do not automatically compose into globally correct authority. Composite repair introduces a higher-level requirement: the subject, artifact, lineage, boundary, target, provenance, and temporal evidence accepted by the local controls must remain jointly applicable to the same protected execution.

8.13 Positive Reachability as a Repair Constraint

The goal is not to transform every AUTHORIZED path into DENIED or UNAVAILABLE. A valid repair must reject or suspend invalid authority while preserving legitimate AUTHORIZED execution when all required evidence is present and applicable. Every repair should therefore be paired with adversarial fixtures and positive controls.

8.14 Repair Evidence Ladder

Level	Evidence state
1	Counterexample demonstrated
2	Repair requirement derived
3	Hostile-reviewed design
4	Implementation exists
5	Adversarial regression evidence
6	Positive AUTHORIZED reachability preserved
7	Exact-state hostile review

These levels describe evidence maturity, not security quality, severity, or ranking.

Verified frozen-v1 evidence levels: objective binding - 7; mutation closure - 7; observation completeness - 7; shutdown/successor - 7; commitment replay/equivocation - 7; verifier rollback - 7; source independence - 7; control-plane independence - 7; delegation - 7; human approval - 7; tool/execution-target - 7; T3 convergence - 7; Composite continuity - 7. These levels record the furthest completed evidence step for each class and must not be interpreted as comparative security scores.

8.15 Section Synthesis

All authority-bearing evidence consumed by an effect must converge on the same protected execution event.

This does not guarantee that admitted evidence is true, complete, or uncompromised. It establishes a narrower property: the system may not justify a present effect using authority evidence that belongs to a different subject, artifact, lineage, boundary, target, history, or execution event. Ground Truth treats that property as authority continuity.

9. Multi-Agent Delegation

Delegation allows authority to cross subjects. It therefore exposes the difference between valid local edges and valid terminal authority. A chain $P \rightarrow A \rightarrow B$ is not sufficient merely because both edges are authentic; the terminal action must remain justified by the authority root and by every required constraint preserved along the chain.

9.1 Grant Identity and Scope

Delegation should preserve the identity of the root grant, allowed re-delegation, objective or artifact scope, subject lineage, boundary context, and temporal applicability. A downstream grant cannot manufacture authority absent from the upstream grant. Re-delegation is itself a permission and must not be inferred from mere ability to act.

9.2 Authority Laundering

Authority laundering occurs when an action that would not be authorized directly becomes apparently authorized after one or more transformations. Scope expansion, objective substitution, artifact substitution, boundary change, subject replacement, temporal drift, or loss of grant identity can each create this condition while every local record remains authentic.

9.3 Temporal Delegation

A delegation valid through event 2 does not authorize an effect at event 3 simply because the delegation record remains the subsystem's most recent valid head. Terminal authorization requires applicability to the event consuming the delegated authority.

9.4 Positive Control

A mature delegation repair must preserve a legitimate chain in which root grant, re-delegation permission, scope, subject lineage, boundary, and temporal applicability all reach the terminal effect.

Reviewed delegation evidence reached Level 7. Hostile cases covered multi-agent delegation authority laundering, stale or transferred grant lineage, scope expansion, and execution-time applicability. The implemented repair requires the exact current grant path, scope, boundary, and effect relationship to remain established; missing continuity yields UNAVAILABLE and established revocation/prohibition yields DENIED. Positive controls preserve fresh, properly scoped delegation as AUTHORIZED.

10. Human Approval Continuity

Human approval is often treated as a high-confidence authorization signal. That confidence can obscure a continuity problem: the fact that a person genuinely approved something does not establish that the person approved the effect now being executed.

"A human approved something" is not the same as "a human approved this effect, by this actor, under this authority context, at this execution event."

10.1 Approval as Relational Evidence

Approval can need to bind to the approving principal, subject, objective, artifact, allowed effect, boundary, execution target, validity interval, and execution event. The precise set depends on the model and implementation; the essential requirement is that approval remain attributable to the protected effect actually being authorized.

10.2 Replay and Transfer

An authentic approval can become stale through replay or can be transferred across a different objective, artifact, agent, connector, account, or effect. Authenticity establishes origin; it does not establish present applicability. A stored approval object must therefore not operate as a bearer token for arbitrary later effects unless the policy explicitly defines it that way.

10.3 Approval and Effect Identity

Autonomous workflows often transform high-level intent into lower-level operations. Approval of an objective can be followed by planning, tool selection, connector resolution, backend selection, and terminal resource mutation. If those transformations materially change the protected effect, approval continuity must be re-established or the result must become DENIED or UNAVAILABLE according to the model.

10.4 Temporal Approval

An approval valid only through event 2 cannot justify a protected effect at event 3. A subsystem may still correctly report historical validity; the composition fails when historical validity is confused with applicability to the current authority-consuming event.

10.5 Human-in-the-Loop Implication

Human involvement does not remove the need for precise authority binding. Ambiguous approval can become more dangerous when downstream components over-trust its mere presence. Ground Truth therefore treats human approval as authority evidence that must be scoped and consumed, not as a universal override.

10.6 Evidence Required

Reviewed human-approval continuity evidence reached Level 7. Hostile cases covered replay, transfer to a different effect or subject, approval aging, revocation, and approval that remained locally valid but did not apply to the terminal execution event. The repair binds approval identity and scope to the required effect/event context, preserves DENIED for established revocation, and uses UNAVAILABLE when required current applicability cannot be established. Fresh reapproval remains a positive AUTHORIZED path.

11. Tool and Execution-Target Continuity

Autonomous systems commonly authorize a logical tool while execution passes through several layers: tool abstraction -> connector -> backend or service -> credential or account -> terminal target. Authority can remain apparently valid at the logical layer while the real effect target changes below it.

11.1 Logical Tool Identity Is Insufficient

A tool name can remain constant while endpoint, connector, account, credential, permission set, region, tenant, or backend changes. If those properties are authority-relevant, authorization bound only to the logical tool is too weak.

11.2 Credential Continuity

A credential can remain technically valid after the authority for using it in the present context has changed. Credential possession establishes capability to authenticate; it does not by itself establish that the current subject, objective, boundary, and effect are authorized.

11.3 Permission Expansion and Backend Substitution

A connector can acquire broader permissions, or a logical operation can resolve to a backend with materially different effects. The authority system should bind the decision to the terminal effect identity required by policy and treat unresolved target identity as UNAVAILABLE rather than assuming equivalence.

11.4 Effect-Binding Chain

A robust authority chain should connect the upstream authorization object to the terminal effect actually executed. Intermediate components can remain implementation details only if they cannot alter authority-relevant meaning. Where they can, their identity or state becomes part of the evidence needed to establish the frozen model.

11.5 Evidence Required

Reviewed tool/execution-target continuity evidence reached Level 7. Hostile cases covered logical-tool stability across connector substitution, backend change, credential hopping, permission expansion, and terminal-target change. The repair requires authority-relevant target and credential continuity through the protected execution. Exact authorized target continuity remains positively reachable; changed or unestablished target state fails closed.

12. Temporal Authority Convergence

Temporal authority convergence is the principal synthesis produced by the long-horizon counterexamples. It addresses a failure that ordinary freshness checks can miss: independently current authority components can still describe different execution histories.

12.1 Event Sequence

Consider an execution history $T1 \rightarrow T2 \rightarrow T3$. Human approval and delegation are valid only through $T2$. A protected tool effect occurs at $T3$. Each subsystem can retain its most recent internally valid record. If authorization asks only whether those records are locally current, the composite system can authorize $T3$ using authority that terminates at $T2$.

12.2 The Join Error

The defect is not necessarily a false local statement. Approval can truthfully report that an approval was valid. Delegation can truthfully report that a grant was valid. A verifier can truthfully report its local head. The invalid step is the join that treats those statements as authority for an execution to which they do not apply.

Local currentness does not imply globally current authority.

12.3 Canonical T3 Execution Event

The implemented repair introduces one protected canonical `T3_EXECUTION_EVENT` as the common authority-consuming join point. Human approval, delegation, tool/connector state, policy, successor/boundary state, verifier state, anchors, witnesses, control-plane evidence, and the terminal effect bind or otherwise demonstrate applicability to that event where policy requires them.

`T3_EXECUTION_EVENT` does not create a seventh public tuple coordinate. It is evidence used to establish that the frozen authority requirements apply to the protected effect.

12.4 Not Continuous Authorization

Authority components can be evaluated asynchronously or cached. Common-event convergence requires only that, at the point authority is consumed, the system can establish that the mandatory evidence applies to the same protected execution.

12.5 Current Evidence Status

The temporal finding progressed from a reproduced long-horizon counterexample to a reviewed common-event design and then to executable repair evidence. The repair does not assert continuous authorization; it establishes the applicability of required authority evidence at the protected execution event.

Exact T3 checkpoint: branch fix/authority-lab-v1-t3-execution-event-convergence; HEAD 4e1eae61136fdda078c5e38f71e0f62f83fa92f9; parent 74fba1f8fd537afb59996662affe1df64457d530; new fixtures LAB-V1-412..443 (32); final T3 fixture count 443; 443/443 Authority Lab fixtures PASS; 257/257 public tests PASS; focused tests 7/7 PASS; zero prior expected-outcome changes. Verdict: PASS — NO REPRODUCIBLE IN-SCOPE CROSS-LAYER TEMPORAL-COMPOSITION OR STALE-EXECUTION BYPASS FOUND.

13. Composite Adversarial Control Continuity

Repairing individual authority modules is insufficient if their composition remains vulnerable. Composite testing asks whether individually acceptable controls are jointly applicable to the same terminal effect.

13.1 Locally Correct, Globally Invalid

A system can combine valid subject identity, valid approval, valid delegation, valid verifier state, valid connector state, and valid witnesses into an invalid authorization if those facts refer to different artifacts, boundaries, roots, targets, histories, or execution events. Composition therefore has its own attack surface.

13.2 Cross-Attack Synthesis

Composite fixtures should intentionally combine repaired classes: successor transitions with delegation, rollback with replay, approval with target substitution, source dependence with witness quorum, and temporal drift across multiple authority heads. Passing unit fixtures for each class is weaker evidence than surviving their adversarial composition.

13.3 Residual Correlation

Composite systems are particularly vulnerable to false independence. Multiple controls can fail together because they share the same source, clock, database, identity root, or control plane. The test corpus should therefore represent known dependency relationships and fail closed when required independence cannot be established.

13.4 Evidence Status

Composite closure reached Level 7 on branch test/authority-lab-v1-composite-adversarial-continuity. Final HEAD 31d78aa158618ca8b72eb61e2f49b434e4319e9e (parent 406a37b4a0019a9eb9d8526b5d248dd1ebc6de20). LAB-V1-444..466 added 23 composite fixtures, bringing the corpus to 466. Public tests passed 259/259 at the Composite checkpoint; Authority Lab fixtures passed 466/466; focused Composite tests passed 2/2; relevant-family regression passed 7/7; eight direct hostile mutations failed closed; no executable Authority Lab code was changed by the Composite slice. Positive paths included fresh composite authority and recovery with current witness continuity. Verdict: PASS — NO REPRODUCIBLE IN-SCOPE COMPOSITE AUTHORITY-CONTINUITY COUNTEREXAMPLE FOUND.

14. Experimental Results

The final reviewed freeze preserves the distinction between experimental evidence, normative requirements, repair evidence, and external assumptions. All numerical results below refer to exact Git states in the remotely inspectable reviewed v1 history. The freeze documentation commit records the frozen research HEAD separately so publication metadata is not conflated with the research result it describes.

14.1 Reviewed Freeze State

- Frozen contract: schema authority-lab-v1; exactly 6 tuple coordinates; exactly 6 invariants; exactly 3 outcomes.
- Frozen corpus: LAB-V0-001..012 and LAB-V1-013..466; 466 total Authority Lab fixtures.

- Frozen public regression suite: 260/260 PASS. Frozen Authority Lab corpus: 466/466 PASS.
- Oracle independence: wrong expectations do not change the actual oracle verdict; CASE_MISMATCH records disagreement.
- Determinism: identical canonical composite-positive input reproduced identical verdict/reason across five runs; targeted independence/determinism run 3/3 PASS.
- T3 checkpoint: 4e1eae61136fdda078c5e38f71e0f62f83fa92f9; 443/443 fixtures; 257/257 public tests; focused 7/7 PASS.
- Composite checkpoint: 31d78aa158618ca8b72eb61e2f49b434e4319e9e; 466/466 fixtures; 259/259 public tests; focused 2/2; relevant-family 7/7; hostile mutations 8/8 failed closed.
- Research freeze: documentation commit 3d09acc804bc1271965696f2d9561143ca28b835 records frozen research HEAD 31d78aa158618ca8b72eb61e2f49b434e4319e9e. A test-only deterministic replay commit precedes the documentation freeze; no executable authority-model semantics changed during the freeze documentation step.

14.2 Results Table

Failure class	Original observed	Normative required	Repaired observed	Evidence level	Exact artifact
Objective binding	Candidate evidence/contract could be correct for the wrong objective or context.	Exact current authoritative objective-to-contract binding required.	Wrong or unestablished binding fails closed; valid bound path remains AUTHORIZED.	7	Freeze evidence manifest GT-PAPER-005
Mutation-registry closure	Authority-relevant mutation could occur outside the observed mutation surface.	Mutation surface must be closed or uncertainty must fail closed.	Unregistered/ambiguous authority-relevant transition no longer inherits prior authority.	7	Freeze evidence manifest / mutation closure slice
Observation completeness	Incomplete observation could be treated as complete authority state.	Admission/completeness must itself be established.	Missing or unestablished required observation becomes UNAVAILABLE.	7	Freeze evidence manifest GT-PAPER-006
Shutdown/successor ABA	Restored visible identity could inherit predecessor authority after discontinuity.	Successor/restart authority must be path-sensitive, not endpoint-only.	ABA/re-entry without exact continuity fails closed; valid successor path remains reachable.	7	Freeze evidence manifest GT-PAPER-007
Commitment replay/equivocation	Authentic old/conflicting commitments could be reused as current evidence.	Commitments require exact freshness, lineage, and conflict handling.	Replay/equivocation paths fail closed; current committed path remains reachable.	7	Freeze evidence manifest GT-PAPER-008
Verifier rollback/split-	Verifier state could regress or	Current verifier authority must	Rollback/split-brain without	7	Freeze evidence manifest GT-

Failure class	Original observed	Normative required	Repaired observed	Evidence level	Exact artifact
brain	diverge while appearing locally valid.	be anchored to non-rolled-back evidence.	current anchor evidence fails closed.		PAPER-009
Anchor/source independence	Several records could share one stale/common source.	Independence is a provenance/control property, not multiplicity.	Correlated sources cannot substitute for required independence.	7	Freeze evidence manifest GT-PAPER-010
Control-plane independence	Nominally separate controls could share hidden authority/control domain.	Control-domain independence must be established where required.	False independence/collusion represented in-model fails closed.	7	Freeze evidence manifest GT-PAPER-011
Delegation laundering	Authentic local grant edges could compose into wider terminal authority.	Exact grant lineage, scope, boundary, and event applicability required.	Stale/transferred/laundered delegation fails closed; fresh scoped path AUTHORIZED.	7	Freeze evidence manifest GT-PAPER-012
Human approval continuity	Real approval could be replayed/transferred to another effect or later event.	Approval must remain current and applicable to the exact protected effect/event.	Revoked approval DENIED; missing/stale applicability UNAVAILABLE; fresh approval AUTHORIZED.	7	Freeze evidence manifest GT-PAPER-013
Tool/backend continuity	Logical tool could remain constant while connector/backend/credential/target changed.	Authority must bind to terminal authority-relevant execution target.	Substitution fails closed; exact current target remains AUTHORIZED.	7	Freeze evidence manifest GT-PAPER-014
T3 convergence	Individually current authority heads could refer to different execution events.	Required authority evidence must converge on one protected T3 execution event.	Stale/mismatched event paths fail closed; fresh converged path AUTHORIZED.	7	Exact head: Section 12.5
Composite continuity	Individually repaired controls could compose into an invalid terminal authorization.	Joint applicability must survive adversarial composition.	No reproducible in-scope composite bypass found at reviewed exact state; positive composite paths preserved.	7	Exact head: Section 13.4

14.3 Interpretation

The results should be interpreted as a record of falsification pressure, not a leaderboard. A mature Level 7 repair means that a reproduced class progressed through exact-state hostile review while retaining positive reachability; it does not mean that the class is universally solved or that the system is secure against unknown variants.

14.4 Reporting Original and Repaired Behavior

The final results table preserves three distinct facts: original observed behavior, normatively required behavior, and repaired observed behavior. This prevents the final passing state from erasing the scientific value of the historical counterexample.

15. Related Work

Ground Truth sits at the intersection of authorization, usage control, delegation, provenance, distributed consistency, rollback resistance, and zero-trust architecture. It should therefore be positioned as a narrow compositional falsification framework rather than as the invention of continuing authorization or context-sensitive access control.

15.1 Zero Trust

NIST SP 800-207 rejects implicit trust based on network location and emphasizes authentication and authorization around users, devices, and protected resources [1]. Ground Truth is compatible with that direction but studies a narrower failure: whether the authority evidence consumed by a terminal effect remains jointly applicable after multi-step transitions. Common-event convergence should therefore not be presented as the invention of continuous verification.

15.2 Usage Control and Ongoing Authorization

The UCONABC model broadened access control to include ongoing controls, mutability, obligations, and conditions, recognizing that authorization for long-lived use can change after initial access [2]. Ground Truth shares the concern that authority is not necessarily a one-time decision. Its distinct emphasis is deterministic counterexample construction over evidence continuity and cross-layer composition at a protected execution event.

15.3 Consistent Authorization State

Google Zanzibar provides a large-scale authorization system whose decisions respect causal ordering amid changes to ACLs and object contents [3]. This work is directly relevant to stale authorization and consistency. Ground Truth is not a replacement for globally consistent authorization storage; it instead asks whether all evidence contributing to a protected autonomous execution refers to the authority context required for that effect.

15.4 Delegation and Attenuated Credentials

Macaroons provide decentralized authorization credentials with contextual caveats and attenuation, allowing authority to be restricted by context, principal, location, or purpose [4]. SPKI/SDSI similarly supports authorization and multi-stage delegation through certificate chains [5]. These systems demonstrate that delegation and contextual restriction are established areas. Ground Truth's delegation contribution is the falsification of continuity failures when authentic local grant edges do not compose into justified terminal authority.

15.5 Fine-Grained OAuth Authorization

OAuth 2.0 Rich Authorization Requests adds structured `authorization_details` so clients can express fine-grained authorization data beyond a flat scope string [6]. Ground Truth is complementary: detailed authorization intent still has to remain bound to the subject, effect, target, boundary, and execution event where it is consumed.

15.6 Provenance and Supply-Chain Integrity

in-toto verifies the integrity of a software supply chain across multiple actors and steps [7], while The Update Framework addresses rollback, freeze, mix-and-match, compromise, delegated trust, and expiration in software

update systems [8]. These systems are useful analogues for lineage, rollback resistance, and explicit trust boundaries. Ground Truth applies similar falsification pressure to authority evidence in autonomous execution rather than to software artifacts alone.

15.7 Contemporary Agent-Security Work

Recent work further sharpens the composition problem. CONTINUITY studies authenticated security-context contracts across composed LLM-agent controls and effect-bound execution permits [9]. Exact checking for checkpoint, fork, restore, and merge operations formalizes what execution-history edits must preserve for safe continuation [10]. Experimental work on shutdown sabotage in multi-agent systems motivates defensive treatment of shutdown and successor transitions without implying that any specific deployed system is rogue [11]. A 2026 survey of authorization architectures for tool-using agents emphasizes identity, delegation, runtime enforcement, prompt-injection bypass, provenance, and accountability [12]. Ground Truth should therefore be read as a narrow deterministic falsification framework for authority continuity, not as a claim that authorization continuity or compositional agent security are unexplored fields.

16. Limitations

16.1 Model-Bounded Results

Authority Lab can only falsify behavior represented by its model and fixture corpus. Unknown authority-relevant dimensions can remain outside the test surface. A PASS therefore has no meaning beyond the declared model, assumptions, and tested mutations.

16.2 Consistent False Evidence

The lab cannot independently recover external truth when all admitted sources lie consistently and no contradictory or independently grounded evidence survives. Provenance checks can expose known common roots; they cannot guarantee the absence of hidden collusion.

16.3 Canonicalization and Oracle Trust

Canonicalization, parser behavior, oracle implementation, and any cryptographic or identity primitives relied upon by the experiment belong to the trusted computing base. A defect in those mechanisms can produce deterministic but incorrect results.

16.4 Fixture Abstraction

Synthetic fixtures isolate authority defects, but they are abstractions of production systems. A minimal reproducer can demonstrate a logical failure without proving that a particular production architecture is exploitable in the same way.

16.5 No Universal Safety Claim

AUTHORIZED does not mean safe, beneficial, correct, or truthful. Ground Truth evaluates authority evidence under a stated model; it does not determine whether the authorized objective is socially desirable or whether an autonomous plan is semantically correct.

16.6 Availability Trade-offs

Fail-closed UNAVAILABLE behavior can reduce availability, especially under partial failures or evidence delays. The paper treats availability and justified authority as separate properties and does not prescribe a universal policy for resolving that trade-off.

16.7 Temporal Event Definition

Common-event convergence depends on a defensible definition of the protected execution event. Some effects are atomic and easy to identify; others unfold over time or across systems. The current T3 design must therefore be

understood as an explicit model boundary, not proof that every real-world effect has an obvious single event identity.

16.8 Reproducibility and Exact-State Publication

The reviewed v1 Git history, including the T3, Composite, deterministic-replay, and research-freeze checkpoints, is remotely inspectable in the public repository. Reproduction should therefore identify the exact commit under test rather than infer results from branch names or later documentation. The freeze documentation commit is 3d09acc804bc1271965696f2d9561143ca28b835; it records frozen research HEAD 31d78aa158618ca8b72eb61e2f49b434e4319e9e. Remote inspectability improves independent reproduction logistics but does not widen the meaning of PASS or alter the distinction among AUTHORIZED, DENIED, and UNAVAILABLE.

17. Discussion

17.1 Why Authority Continuity Matters for Autonomous Systems

Autonomous systems can retain state and act across long intervals without renewed human attention. A human approval, delegated task, connector session, or queued job can outlive the authority context in which it was created. As workflows become more distributed and asynchronous, the distance between initial intent and terminal effect can grow.

17.2 The Problem Is Not Merely Staleness

Staleness is only one manifestation. A fact can be fresh but applicable to the wrong subject, artifact, grant lineage, boundary, target, or event. Ground Truth therefore treats authority continuity as a binding and composition problem, not merely a timestamp problem.

17.3 DENIED and UNAVAILABLE as Separate Engineering States

Preserving UNAVAILABLE changes system design. A binary allow/deny API pressures missing information into one side. Three-way authority lets the system distinguish known impermissibility from inability to establish the evidence required for a decision. This distinction is especially useful in distributed systems where uncertainty is ordinary rather than exceptional.

17.4 Authority Evidence as a First-Class Artifact

One architectural implication is that authority evidence should be treated as a first-class artifact with identity, provenance, lineage, applicability, and consumption semantics. Systems that preserve only the final Boolean decision discard information needed to determine whether that decision remains valid after subsequent transitions.

17.5 Research Utility of Deterministic Counterexamples

The methodology converts architectural arguments into inspectable artifacts. A reviewer can disagree with the normative rule, fixture construction, oracle, or repair; each disagreement is localized. The value of the approach is not certainty, but inspectability and reproducibility.

17.6 Toward Execution-Centered Authorization

Common-event convergence suggests an execution-centered view of autonomous authorization. Instead of asking only whether each component possesses a valid local authorization state, the terminal decision asks whether the required evidence collectively justifies this effect now. Cached, delegated, asynchronous, and distributed controls can still be used if their applicability is demonstrated at consumption.

17.7 Future Work

- Replicate the frozen corpus and hostile-review process independently from the exact remotely inspectable release state.

- Study distributed definitions of a protected execution event for non-atomic and multi-stage effects.
- Evaluate authority continuity in real agent frameworks, tool routers, connectors, and execution sandboxes without weakening the public/private disclosure boundary.
- Evaluate runtime authority enforcement separately from the frozen research corpus, including protected side effects, durable authority state, and backend isolation.
- Explore machine-readable evidence manifests that map terminal effects to the exact authority facts consumed.
- Investigate formal models and mechanized proofs that complement, rather than replace, deterministic falsification.
- Archive release artifacts with immutable paper-to-fixture mappings and, where useful, DOI/preprint identifiers.

18. Conclusion

Autonomous systems can preserve the machinery of action more reliably than they preserve the justification for action. Credentials persist. Approvals persist. Delegations persist. Tools remain callable. Workers restart. Queues retry. Control planes rebuild state. None of those facts alone proves that the protected effect remains authorized.

Ground Truth frames this as authority continuity and tests it through deterministic falsification. Its core distinction is simple: capability, evidence, authority, and execution are different properties. The presence of one must not be treated as proof of another.

The counterexamples developed in this manuscript converge on a compact architecture of canonical binding, closure, lineage, non-rollbackable authority-relevant history, provenance, fail-closed uncertainty, effect identity, boundary continuity, anti-replay, durable consumption, and common-event convergence. They therefore describe recurring ways authority can be inherited after its evidence has ceased to apply rather than a collection of unrelated bugs.

The temporal result sharpens the problem. Local authority heads can each be internally current and still fail to justify the same execution. An approval valid through event 2 and a delegation valid through event 2 do not become authority for a protected effect at event 3 simply because their records remain present and locally valid.

Local currentness does not imply globally current authority.

The resulting requirement is narrower than continuous authorization and stronger than independent freshness: when a protected effect consumes authority, the evidence required by policy must converge on the same authority-consuming execution event.

Authority continuity is not persistence of authorization; it is the persistence or re-establishment of the evidence required to justify authorization.

Ground Truth does not claim universal security. It does not establish the truth of all external evidence. It does not prove the absence of unknown counterexamples. Its goal is more concrete: make authority failures reproducible, make repairs inspectable, preserve legitimate authorization, bind claims to exact executable evidence, and make PASS mean only what the experiment can actually support.

Reproducibility Package - Publication Checklist

- Canonical public repository and frozen authority-lab-v1 specification.
- Verbatim six-invariant definitions and tuple semantics.
- Exact release commit/tag and clean state used for paper results.
- Full fixture corpus and canonical schema/canonicalization rules.
- Oracle implementation and oracle-independence test.
- Exact commands and runtime/environment versions.
- Results output archived with hashes where useful.
- Paper-to-fixture and paper-to-review-artifact mapping.
- Positive AUTHORIZED, DENIED, and UNAVAILABLE canonical controls.
- T3 and composite evidence status stated independently of completed repair classes.

- Archive and DOI/preprint identifiers are publication metadata, not experimental evidence; they may be added without changing the frozen research result.

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Appendix A. Canonical Public Contract

Canonical six-invariant text is reproduced verbatim in Section 2.1 from LAYER2_COMMIT_TIME_AUTHORITY_REVIEW.md. The frozen tuple is (subject, artifact, control_state, identity_basis, boundary_epoch, decision). The canonical invariant identifiers are INV-CONT, INV-DISC, INV-BND, INV-EVD, INV-USE, and INV-CLO.

Appendix B. Evidence Manifest

Paper claim	Evidence category	Fixture/test	Exact state	Artifact/review	Status
GT-PAPER-001	Normative + experimental framing	README / model	Frozen v1	Canonical contract	Verified
GT-PAPER-002	Experimental/ model	Outcome tests	Frozen v1	AUTHORIZED / DENIED / UNAVAILABLE	Verified
GT-PAPER-003	Experimental	Oracle-independence	Freeze	AuthorityLabTests	Verified

Paper claim	Evidence category	Fixture/test	Exact state	Artifact/review	Status
		tests			
GT-PAPER-004	Experimental	Repeated deterministic replay	Freeze	test_identical_canonical_input_replays_deterministically	Verified
GT-PAPER-005	Repair	Objective binding family	Freeze	Paper evidence manifest	Level 7
GT-PAPER-007	Repair	Shutdown/successor family	Freeze	Paper evidence manifest	Level 7
GT-PAPER-012	Repair	Delegation family	Freeze	Paper evidence manifest	Level 7
GT-PAPER-013	Repair	Human approval family	Freeze	Paper evidence manifest	Level 7
GT-PAPER-014	Repair	Tool/target family	Freeze	Paper evidence manifest	Level 7
GT-PAPER-015	Experimental	Long-horizon temporal counterexample	15780ddb8d7cb864e0170eb5fab489451daa5bd9	Temporal composition research	Remotely inspectable
GT-PAPER-016	Repair	LAB-V1-412..443	4e1eae61136fdda078c5e38f71e0f62f83fa92f9	T3 execution-event convergence	Level 7
GT-PAPER-017	Repair	LAB-V1-444..466	31d78aa158618ca8b72eb61e2f49b434e4319e9e	Composite adversarial continuity	Level 7
GT-PAPER-018	Experimental	Positive controls across repaired classes	Freeze	Paper evidence manifest	Verified
GT-PAPER-019	Experimental/model	DENIED/UNAVAILABLE controls	Freeze	Oracle + fixtures	Verified
GT-PAPER-020	External assumption	Consistent-lie boundary	Freeze	Threat model / limitations	Explicit limitation